Homework #9

This is the homework for week 10. The due date is 11:59 p.m., November 26. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (35 points) Suppose data $X_1, \cdots, X_n$ are i.i.d. and satisfy Poission distribution, i.e.,

$$Prob(X_i = k) = \frac{\lambda^k}{k!} \exp(-\lambda).$$

Suppose we want to test the hypothesis

$$H_0 : \lambda = \lambda_0 \text{ vs } H_1 : \lambda \neq \lambda_0.$$

i. Suppose we want to use likelihood ratio test. Derive the formula of the test statistics and the p-value.

ii. Suppose we want to use Wald test. Derive the formula of the test statistics and the p-value.

iii. Suppose we want to use Score test. Derive the formula of the test statistics and the p-value.

iv. Suppose we have data set $[4, 1, 1, 2, 1, 3, 1, 3, 4, 3, 4, 0, 2, 2, 2, 1, 0, 1, 2, 3]$, and $\lambda_0 = 1.0$. For these three tests, with size (Type-I error) 5%, do we reject H_0?

(**Question2):** (35 points) Let $X_1, \cdots, X_n$ be n independent random variables with normal distributions $X_i \sim N(x_i\beta, \sigma^2)$, where x_i and σ^2 are known. We want to test the hypothesis

$$H_0 : \beta = \beta_0 \; vs \; H_1 : \beta \neq \beta_0.$$

i.Suppose we want to use likelihood ratio test. Derive the formula of test statistics and the P-value of the test.

ii. Suppose we want to use Wald test. Derive the formula of test statistics and the P-value of the test.

2 Programming Questions

(30 points) In this assignment, we will investigate the performance of P-value under H_0 and H_1. Let us suppose observations $X_1, \cdots, X_n \sim N(\mu, \sigma^2)$, with σ^2 known. Suppose we want to test the hypothesis

$$H_0 : \mu = \mu_0 \text{ versus } H_1 : \mu > \mu_0.$$

i. Suppose $\mu = \mu_0 = 1.0$, $\sigma^2 = 2$, $n = 100$. Write down the formula of P-value.

ii. Generate a set of data $X_1, \cdots, X_{100} \sim N(1, 2)$. Calculate the P-value p_i using your formula. Repeat this for 500 times, and derive the p-values $p_1, \cdots, p_{500}$. Plot a histogram of p_i, and it should look like uniform distribution.

iii. Now suppose $\mu = 1.0$ and $\mu_0 = -1.0$. Again calculate the P-value p_i and repeat for 500 times. Plot a histogram of p_i, and the p-value should concentrate on 0.